Light

GCSE Computer Science — Cyber Security Prevention (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Software securely stores all a user's passwords in an encrypted vault, requiring only one master password to access them.
What is this software called?

- **A.** A password manager
- **B.** A web browser
- **C.** A spreadsheet
- **D.** An operating system

Q2 (1 mark)

A computer is configured to download and install security patches automatically without user intervention.
What is the main security benefit of this configuration?

- ○ **A.** It makes the computer run faster
- ○ **B.** Vulnerabilities are patched promptly, reducing the window of opportunity for attackers to exploit known flaws
- ○ **C.** It increases available storage space
- ○ **D.** It improves the display resolution

Q3 (1 mark)

Software automatically scans incoming emails and moves messages identified as spam or containing malware to a quarantine folder.
What is this security measure called?

- ○ **A.** Data compression
- ○ **B.** File defragmentation
- ○ **C.** Email filtering
- ○ **D.** Password hashing

Q4 (1 mark)

A smartphone automatically locks its screen and requires a PIN or fingerprint after 30 seconds of inactivity.
What security benefit does this provide?

- ○ **A.** It improves battery life
- ○ **B.** It makes the phone lighter
- ○ **C.** It increases storage capacity
- ○ **D.** It prevents unauthorised access if the device is lost or left

unattended

Q5 (1 mark)

A website URL begins with 'https://' rather than 'http://'.
What does the 's' indicate about the connection?

- ○ **A.** The connection is encrypted using TLS/SSL
- ○ **B.** The website loads faster
- ○ **C.** The website contains no advertisements
- ○ **D.** The website is free to use

Q6 (1 mark)

Anti-malware software continuously monitors file activity and scans files as they are opened or downloaded.
What is this feature called?

- ○ **A.** Scheduled scanning
- ○ **B.** Disk defragmentation
- ○ **C.** File compression
- ○ **D.** Real-time protection

Q7 (1 mark)

A social media user configures their account so only approved friends can see their posts and personal information.
What type of security control is this?

- ○ **A.** Privacy settings controlling who can view personal information
- ○ **B.** Antivirus protection
- ○ **C.** Firewall configuration
- ○ **D.** Data compression

Q8 (1 mark)

A user's phone automatically uploads photos to an online storage service whenever connected to WiFi.
What type of backup does this provide?

- ○ **A.** Manual local backup
- ○ **B.** Disk defragmentation
- ○ **C.** Automatic cloud backup
- ○ **D.** File compression

Q9 (1 mark)

After receiving a phone call requesting sensitive information, an employee hangs up and calls the organisation back using a number from their official website.
What security practice is this?

- ○ **A.** Data encryption
- ○ **B.** File compression
- ○ **C.** Callback verification to confirm caller identity

- ○ **D.** Password hashing

Q10 (1 mark)

A company uses a cross-cut shredder to destroy paper documents containing customer information before disposal.
Why is this practice important?

- ○ **A.** It makes the paper easier to recycle
- ○ **B.** It reduces the weight of waste bins
- ○ **C.** It prevents dumpster diving attacks where attackers recover sensitive information from rubbish
- ○ **D.** It improves air quality in the office

Q11 (1 mark)

Employees must scan an electronic badge to enter the office building, and each entry is logged with time and identity.
What security benefits does this system provide?

- ○ **A.** It plays music when employees enter
- ○ **B.** It improves the building's WiFi signal
- ○ **C.** It reduces electricity consumption
- ○ **D.** It controls physical access and creates an audit trail of who entered and when

Q12 (1 mark)

A web browser displays a warning that a website's security certificate has expired.
What should the user do?

- ○ **A.** Avoid entering sensitive information as the connection may not be properly secured
- ○ **B.** Ignore the warning and proceed to enter personal information
- ○ **C.** Disable all browser security features
- ○ **D.** Close the browser and never use the internet again

Q13 (2 marks)

A network administrator configures systems to alert when unusual patterns of activity occur, such as large data transfers at night or connections to known malicious IP addresses.
What security capability does this provide?

- ○ **A.** Intrusion detection that identifies potentially malicious activity through anomaly and signature-based monitoring
- ○ **B.** File compression that reduces storage requirements
- ○ **C.** Disk defragmentation that improves performance
- ○ **D.** Password hashing that protects stored credentials

Q14 (2 marks)

A web developer ensures that all user input is checked to only contain expected characters and formats before processing.
What attack does this primarily prevent, and why?

- ○ **A.** Physical theft; input validation locks office doors

- ○ **B.** Input validation prevents injection attacks by ensuring user data cannot contain malicious code that the application might execute
- ○ **C.** Power outages; input validation provides backup electricity
- ○ **D.** Hardware failure; input validation repairs broken components

Q15 (2 marks)

A company's mobile device management system allows administrators to remotely erase all data from a lost or stolen employee phone.
What is this capability called, and why is it important?

- ○ **A.** Remote wipe; it ensures sensitive company data cannot be accessed if devices fall into the wrong hands
- ○ **B.** Remote backup; it copies data to the cloud
- ○ **C.** Remote defragmentation; it improves phone performance
- ○ **D.** Remote compression; it reduces file sizes

Q16 (2 marks)

A system records all login attempts, file access events, and configuration changes with timestamps and user identities.
Why is maintaining these logs important for security?

- ○ **A.** Logs improve system performance by tracking resource usage
- ○ **B.** Logs compress data to save storage space
- ○ **C.** Logs provide audit trails for investigating incidents, detecting patterns, and demonstrating compliance

- ○ **D.** Logs defragment the hard drive automatically

Q17 (2 marks)

A laptop is configured so that the entire hard drive contents are encrypted, requiring a password at startup before the operating system loads.
What protection does this provide?

- ○ **A.** It makes the laptop run faster
- ○ **B.** It increases the laptop's battery life
- ○ **C.** It improves the screen resolution
- ○ **D.** Full disk encryption protects data if the device is lost or stolen — without the password, the drive contents cannot be read even if removed

Q18 (2 marks)

A company requires that the employee who requests a payment is different from the employee who approves it.
What security principle does this implement?

- ○ **A.** Single sign-on for convenience
- ○ **B.** Separation of duties to prevent fraud by requiring multiple people for sensitive actions
- ○ **C.** Open access to improve efficiency
- ○ **D.** Minimal logging to save storage

Q19 (2 marks)

A home user configures their WiFi router to use WPA3 encryption instead of the older WEP standard.
Why is this an important security improvement?

- ○ **A.** WPA3 makes the WiFi signal travel further
- ○ **B.** WPA3 reduces electricity consumption
- ○ **C.** WPA3 provides much stronger encryption that is resistant to attacks that can break WEP within minutes
- ○ **D.** WPA3 improves internet download speeds

Q20 (2 marks)

A system automatically disables a user account for 30 minutes after five consecutive incorrect password attempts.
What attack does this measure primarily defend against?

- ○ **A.** Physical theft of computer equipment
- ○ **B.** Brute force attacks that systematically try password combinations
- ○ **C.** Power outages affecting the data centre
- ○ **D.** Natural disasters destroying backup tapes

Q21 (2 marks)

After installing a new router, an administrator immediately changes the default username and password.
Why is this an essential security step?

- ○ **A.** Default passwords are printed in public documentation that attackers can easily find, providing immediate access to unconfigured devices
- ○ **B.** Changing passwords improves WiFi signal strength
- ○ **C.** New passwords increase internet speed
- ○ **D.** Modified credentials reduce power consumption

Q22 (2 marks)

A company stores data in a cloud service. The cloud provider secures the physical data centres and infrastructure, while the company must secure their own data, access controls, and user accounts.
What security model does this describe?

- ○ **A.** The cloud provider is responsible for all security with no customer involvement
- ○ **B.** The customer is responsible for all security including physical data centres
- ○ **C.** No security is required when using cloud services
- ○ **D.** Shared responsibility where security duties are divided between provider and customer

Q23 (2 marks)

A company disables USB ports on all workstations except for approved, encrypted USB devices.
What threats does this policy address?

- ○ **A.** It improves monitor display quality
- ○ **B.** It prevents data theft via USB drives and blocks malware

introduction through infected removable media

○ **C.** It increases keyboard typing speed

○ **D.** It reduces paper usage in the office

Q24 (3 marks)

A company's disaster recovery plan includes:
• Recovery Time Objective (RTO): 4 hours
• Recovery Point Objective (RPO): 1 hour
• Off-site backup data centre
• Documented recovery procedures
• Annual testing and updates
What do RTO and RPO mean, and why are both important?

○ **A.** RTO and RPO are the same thing — both measure backup storage capacity

○ **B.** RTO measures internet speed; RPO measures disk space

○ **C.** Neither metric is relevant to disaster recovery planning

○ **D.** RTO is maximum acceptable downtime (4 hours to restore services); RPO is maximum acceptable data loss (up to 1 hour of recent work); together they define recovery requirements that determine backup frequency and infrastructure investment

Q25 (3 marks)

A security analyst compares two organisations:
Organisation A: Strong firewall only
Organisation B: Moderate firewall + email filtering + endpoint protection + user training + network monitoring
Despite Organisation A's firewall being technically superior, Organisation B experiences fewer successful attacks.
Which analysis correctly explains this outcome?

○ **A.** Firewall strength is the only factor that matters; Organisation B's success must be coincidental

○ **B.** Organisation B's defence-in-depth approach means attacks blocked by any layer do not succeed; Organisation A's single-layer defence fails completely if bypassed — phishing emails bypass firewalls, so trained users become the defence

○ **C.** Organisation B has fewer employees, explaining the difference

○ **D.** Technical controls are irrelevant; only physical security matters

Q26 (3 marks)

An organisation prepares for security incidents by:

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• Documenting response procedures for common incident types
• Assigning roles and responsibilities to team members
• Establishing communication channels and escalation paths
• Creating relationships with external resources (forensics, lega
• Conducting regular tabletop exercises
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Why is this preparation essential rather than simply responding to incidents as they occur?

○ **A.** Prepared responses are faster, more thorough, and less error-prone; during actual incidents, stress and time pressure cause mistakes if teams must figure out procedures on the spot; practice builds muscle memory and identifies gaps before real incidents

○ **B.** Preparation provides no benefit; improvised responses are equally effective

○ **C.** Documentation only helps with compliance, not actual incident handling

○ **D.** External relationships are unnecessary; all incidents should be handled internally

Q27 (3 marks)

A hospital implements access controls where:
• Doctors can view and modify their own patients' records

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• Nurses can view records for patients on their ward
• Administrators can manage user accounts but cannot view patient
• Researchers can access anonymised statistical data only
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What access control approach is this, and why is it appropriate for healthcare?

- ○ **A.** Everyone should have full access to improve collaboration
- ○ **B.** Role-based access control (RBAC) assigns permissions based on job function; it is appropriate because healthcare involves sensitive data requiring strict need-to-know access while enabling staff to perform their duties; different roles need different data access
- ○ **C.** Only doctors should have any system access
- ○ **D.** Access controls are unnecessary in healthcare settings

Q28 (3 marks)

A company's security programme includes:
• Vulnerability scanning: Automated tools check for known vulnerabilities monthly
• Penetration testing: Ethical hackers attempt to exploit systems annually
• Security audits: External auditors review policies and controls annually
Why does the company use all three rather than just one?

- ○ **A.** One method is sufficient; using three is wasteful redundancy
- ○ **B.** These methods are identical and produce the same results
- ○ **C.** Only penetration testing is valuable; other methods are ineffective

○ **D.** Each method serves a different purpose: vulnerability scanning finds known issues continuously at low cost; penetration testing demonstrates real exploitability through skilled attackers; audits assess policies and compliance that technical tests miss — together they provide comprehensive security assessment

Q29 (3 marks)

A company's security awareness programme includes:
• Initial training for all new employees
• Monthly security newsletters with current threat information
• Quarterly simulated phishing exercises with feedback
• Annual refresher training with updated content
Simulated phishing results show click rates dropping from 35% to 8% over two years.
What does this programme demonstrate about security awareness training?

○ **A.** Training has no measurable effect on user behaviour

○ **B.** One-time training is sufficient; ongoing programmes are unnecessary

○ **C.** Continuous, multi-format training measurably improves security behaviour; the combination of initial education, regular reinforcement, practical exercises with feedback, and updated content creates lasting behavioural change that single training sessions cannot achieve

○ **D.** Only technical controls matter; user training is ineffective

Q30 (3 marks)

A business continuity plan addresses:
• Alternative work locations if offices are inaccessible

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• Manual procedures if IT systems are unavailable
• Key supplier contingencies if primary vendors fail
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• Communication plans for staff, customers, and stakeholders
• Priority ranking of business functions for resource allocation
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How does business continuity planning differ from disaster recovery, and why are both needed?

- ○ **A.** Business continuity and disaster recovery are identical concepts
- ○ **B.** Disaster recovery focuses on restoring IT systems; business continuity ensures the organisation can continue operating during and after disruption through alternative processes, locations, and resources — both are needed because IT recovery alone does not guarantee business operations can continue
- ○ **C.** Only IT systems matter; business processes are irrelevant to recovery
- ○ **D.** Planning is unnecessary; organisations should simply respond when disasters occur

Total: **55** marks • Time: **30 min**

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